Algorithm 2 Procedural generation of the botanical portrait

Require: Canvas size $W = 1024$, $H = 576$; fixed seed s
Ensure: Rendered RGB image I

$C \leftarrow \textsc{NewCanvas}(W, H)$
$R_f, R_w, R_p \leftarrow \textsc{SeededGenerators}(s, \text{foliage}, \text{woman}, \text{flowers})$
▷ Layer 1: white background
$\textsc{FillRectangle}(C, (0, 0, W, H), \text{white})$
▷ Layer 2: circular foliage
for $i = 0$ to 25 **do**
    $\theta \leftarrow 2\pi i/26 + \textsc{Jitter}(R_f)$
    $\mathbf{x}, \mathbf{y} \leftarrow \textsc{BranchEndpoints}((W/2, H/2), \theta, R_f)$
    $\textsc{StrokeBezierBranch}(C, \mathbf{x}, \mathbf{y})$
    for each position along the branch **do**
        $\textsc{DrawGradientLeaves}(C, R_f)$
        $\textsc{DrawLeafVeinsAndGrain}(C, R_f)$
    end for
end for
$\textsc{DrawBerrySprigsAndAdditionalLeaves}(C, R_f)$
▷ Layer 3: woman's portrait
$\textsc{FillBezierShape}(C, \text{hair silhouette}, \text{brown gradient})$
$\textsc{FillBezierShape}(C, \text{neck and shoulders}, \text{skin gradient})$
$\textsc{FillBezierShape}(C, \text{face}, \text{skin gradient})$
$\textsc{AddClippedShadingAndGrain}(C, \text{face}, R_w)$
$\textsc{DrawFacialPaths}(C, \text{brows, closed eyes, lashes, nose, lips})$
$\textsc{FillBezierShape}(C, \text{front hair locks}, \text{brown gradient})$
for $d \in \{\text{left}, \text{right}\}$ **do**
    for $j = 0$ to 74 **do**
        $\textsc{StrokeClippedHairStrand}(C, d, j, R_w)$
    end for
end for
▷ Layer 4: flowers and foreground details
for each prescribed rose position (x, y, r) **do**
    $\textsc{DrawSepals}(C, x, y, r)$
    for $k = 0$ to 4 **do**
        $n \leftarrow [8, 7, 6, 5, 4]_k$
        for $j = 0$ to $n - 1$ **do**
            $\textsc{DrawRotatedBezierPetal}(C, x, y, r, k, j, R_p)$
            $\textsc{AddPetalGradientAndGrain}(C, R_p)$
        end for
    end for
end for
for each prescribed daisy or small-blossom position **do**
    $\textsc{DrawRadialPetalsAndCenter}(C, R_p)$
end for
$\textsc{DrawForegroundLeavesAndBerries}(C, R_p)$
$I \leftarrow \textsc{Rasterize}(C)$
return I

6.3 Comparative Plots and Table Conventions

Radar rankings. Fig. 2 uses six metrics from Tables 2 and 3: mean alignment, aesthetics, and composition scores; generation success rate; reported time per task; and the sum of input and output tokens per task. Each metric is ranked across all 11 models before grouping them into panels. Higher quality and success values rank better, whereas lower costs rank better. Ties in the reported values receive average ranks. A rank r is mapped to radius $(12 - r)/11$, placing rank 1 at the outer edge. Polygon area is not an aggregate performance score, and radial differences represent rank differences rather than differences in the underlying metrics. Legend counts indicate successful